DATA ANALYST INTERN ASSIGNMENT

Mini Project: Retail Sales Data Analysis Using Python

Company: Infinite Business Ventures (Pvt) Ltd
Suggested duration: 2–3 days | **Difficulty:** Beginner–Intermediate

1. Project objective

Use Python to clean and analyze a retail sales dataset, create meaningful visualizations, and produce an Excel summary that can support business decisions.

2. Dataset provided

Use **Retail_Sales_Intern_Dataset.xlsx**. It contains Raw_Sales_Data and Data_Dictionary sheets. The raw data intentionally contains data-quality issues. Do not edit the raw sheet directly; load it into Python and create a cleaned copy.

Task 1 — Inspect and clean the data (25 marks)

• Load the Raw_Sales_Data sheet using pandas.

• Show row/column counts, data types, missing values, and duplicate records.

• Handle missing values, duplicate rows, invalid quantities/prices/discounts, unknown brands/outlets, and inconsistent brand text. Explain each decision.

• Validate Net_Sales_LKR against Quantity × Unit_Price_LKR × (1 – Discount_Rate). Flag discrepancies rather than silently changing them.

• Keep the raw data unchanged and save a cleaned dataset separately.

Task 2 — Analyze sales (25 marks)

• Calculate total net sales, units sold, invoice count, and average transaction value (ATV). Define how you calculate invoice count and ATV.

• Create brand-wise, outlet-wise, and month-wise sales summaries.

• Identify the top five brands by net sales and top five products by net sales.

• Compare discount rates and sales performance; describe at least one useful pattern.

Task 3 — Visualize the results (15 marks)

• Create at least three charts: monthly sales trend, brand-wise net sales, and outlet-wise net sales.

• Give each chart a clear title, axis labels, readable formatting, and LKR units where relevant.

Task 4 — Create an automated Excel report (20 marks)

• Export a cleaned data sheet and summary sheets to a new Excel workbook.

• Include at least: KPI Summary, Brand Summary, Outlet Summary, Monthly Summary, and Cleaned Data.

• Your Python script/notebook must regenerate the report when run again without manually editing the output.

Task 5 — Business insights (10 marks)

• Write five evidence-based findings and three practical recommendations for management.

• Each finding should reference a result from your analysis; avoid unsupported assumptions.

Task 6 — Code and presentation (5 marks)

• Use clear variable names, comments, and reusable functions where appropriate.

• Be ready to explain your cleaning decisions, formulas, and charts in a 5–10 minute walkthrough.

4. Required tools

Python, pandas, matplotlib (or seaborn), openpyxl, and Jupyter Notebook or Google Colab. Internet access is not required.

5. Deliverables

• Python script (.py) or Jupyter Notebook (.ipynb)

• Generated Excel report (.xlsx)

• Three or more chart files or charts embedded in the notebook/report

• One-page management summary (PDF, Word, or in the notebook)

6. Marking rubric

Assessment area	Marks	What is assessed
Data cleaning and validation	25	Correct handling and explanation of data issues
Sales analysis	25	Accurate metrics and useful group summaries
Visualization	15	Clear and appropriate charts
Automated Excel report	20	Required sheets and repeatable output
Business insights	10	Evidence-based findings and recommendations
Code and walkthrough	5	Readable code and understanding
TOTAL	100	

7. Submission and evaluation notes

Submit all files using clear filenames. Do not include personal or confidential data. You may use documentation or AI tools for learning, but disclose their use and be able to explain and modify every part of your solution. Evaluation focuses on accuracy, reasoning, reproducibility, and business communication—not just whether the code runs.

8. Optional challenge (not included in the 100 marks)

Build a simple Streamlit dashboard with date, brand, and outlet filters, or add a data-quality summary showing how many records were corrected, excluded, or flagged.